

The connection between aurorae and variations in Earth's magnetism was discovered by English clockmaker George Graham in 1722.

The Russian Academy of Sciences was founded in St Petersburg in 1724.

WITH SO MANY BUILDINGS MADE MOSTLY OF WOOD,

fire was a major hazard in 18th-century cities. In the 1600s, the Dutch had rushed **water pumps** mounted on handcarts to fires, but they delivered little more than a trickle of water. The breakthrough came when London buttonmaker **Richard Newsham** (d.1743) patented a pump in North America in 1721. Newsham's **fire pump cart** was the forerunner of today's fire engines. It had a 169-gallon (640-liter) watertank, and its pump operated by long handles and foot treadles extending either side could squirt 100 gallons (380 liters) of water per minute.

In 1722, clockmaker George Graham (1674–1751) noticed the link between aurorae (natural light displays in the sky) and **Earth's magnetism**. He observed that magnetic "storms" that made a compass needle fluctuate significantly coincided with sightings of the aurorae. Graham's discovery followed a particularly dramatic display of the **aurora borealis**, or northern lights, in 1716 that had fascinated people at the time and was seen

Accurate timekeeping
The mercury pendulum helped eliminate inaccuracies in timekeeping caused by temperature variations with solid weights.

RENÉ ANTOINE FERCHAULT DE RÉAUMUR (1683–1757)

Born in La Rochelle, France, René Réaumur was a naturalist who made contributions to many different fields of science, from the study of insects to ceramics and metallurgy. Elected to the French *Academie des Sciences* (Academy of Science) aged just 24, his greatest work was in natural history, where he showed that some crustaceans can regenerate lost limbs.



as far south as London. Graham also improved the accuracy of **pendulum clocks** by replacing the solid lead weight with a flask of liquid mercury. This eliminated the variations in the length and

swing of a solid weight caused by expansion and contraction due to temperature change.

In France, polymath **René Réaumur** (1683–1757) was experimenting with **iron and steel**. He realized that the difference between the metals was caused by their differing **sulfur and salt contents**. Steel, produced by smelting iron, was more brittle than pure iron because it contained sulfur, while cast iron was even more brittle because it contained still higher levels of sulfur. Réaumur discovered that the brittleness of cast iron could be reduced by burying it in lime to draw sulfur out. He believed this method was too expensive to be practical but it later became widely used.

620 MILES
THE HEIGHT
OF SOME
NORTHERN
LIGHT
DISPLAYS

IN PARIS IN 1723, Italian astronomer **Giacomo Filippo Maraldi** (1665–1729) noticed that there was a bright spot in the center of the shadow of any disk. This phenomenon, later called the **Arago spot**, is caused by interference between waves of light coming around the edge of the object. Maraldi's observation later became proof of the theory that **light travels in waves**, not particles, because only waves can produce an interference pattern (see 1801).

Also in Paris that year, naturalist **Antoine de Jussieu** (1686–1758) compared stones called **ceraunia**, thought to be natural, to the stone tools of Native Americans. The likeness proved that ceraunia were ancient axes and arrowheads.

In 1724, Russian emperor **Peter the Great** (1672–1725) founded the St. Petersburg Academy of Sciences and installed Swiss mathematician **Daniel Bernoulli**

Prehistoric tool
Originally thought to be of natural origin, ceraunia stones like this arrowhead came to be understood as man-made devices.



1721 London buttonmaker Richard Newsham patents a fire pump in North America

1722 George Graham introduces the mercury pendulum

1722 British clockmaker George Graham shows the link between aurorae and Earth's magnetism

1722 French polymath René Réaumur shows the importance of sulfur content in iron and steel

1723 Italian astronomer Giacomo Maraldi observes the Arago spot